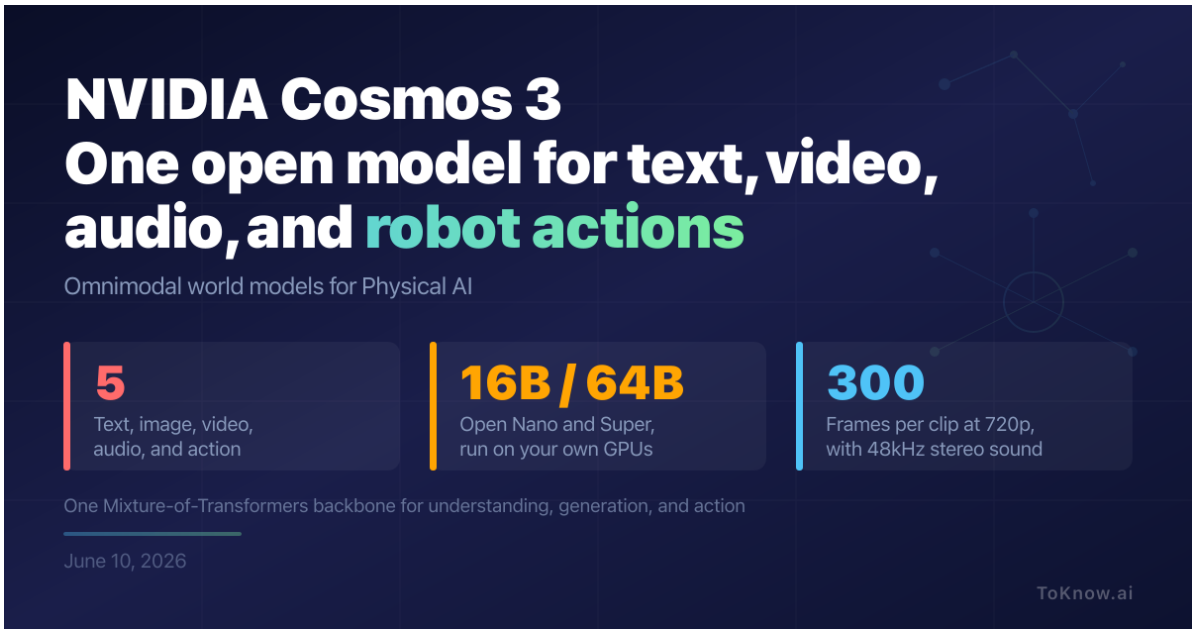
NVIDIA Cosmos 3: One Open Model for Text, Images, Video, Audio, and Robot Actions

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NVIDIA released Cosmos 3, a family of "omnimodal" world models that read and generate five things at once: text, images, video, audio, and robot actions. A single Mixture-of-Transformers design pairs an autoregressive transformer for understanding with a diffusion transformer for generation, so the same weights can describe a scene, imagine how it unfolds, and plan a robot's moves. It comes in a 16B "Nano" and a 64B "Super" size, and can generate clips up to

300 frames at 720p with synchronized 48kHz stereo sound, plus action trajectories for robots and vehicles. NVIDIA reports the models ranked best among open systems for image and video generation, and topped a robot-policy benchmark.

Building a robot or self-driving stack usually means stitching together separate models: one to read the camera, one to predict what comes next, one to pick actions. Cosmos 3 folds all of it into one set of weights you can train together. Because NVIDIA released the code, checkpoints, synthetic training scenes, and benchmark under an open license, a small team can run the 16B model on its own GPUs instead of a closed API, and use it to generate the training data robots usually lack.

This moves world models from research demos toward a shared, open backbone for Physical AI, echoing efforts like OpenWorldLib to unify the many kinds of world model. The open question is no longer whether a model can add a modality, but whether one model can do all of them well enough to trust on a robot.

Sources:

- NVIDIA Cosmos 3 technical report (arXiv)
- NVIDIA/cosmos on GitHub
- Cosmos 3 project website (NVIDIA Research)
- Cosmos 3 model collection on Hugging Face

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